Lab Experiment - 5

Aim

To implement the Porter Stemmer algorithm using the NLTK library in Python to perform word stemming on a list of words.

Algorithm

1. Start the program.
2. Import the `PorterStemmer` class from NLTK.
3. Create a Porter Stemmer object.
4. Define a list of words.
5. Apply the `stem()` method to each word.
6. Display the original words and their stemmed forms.
7. Stop the program.

Python Program

```
from nltk.stem import PorterStemmer

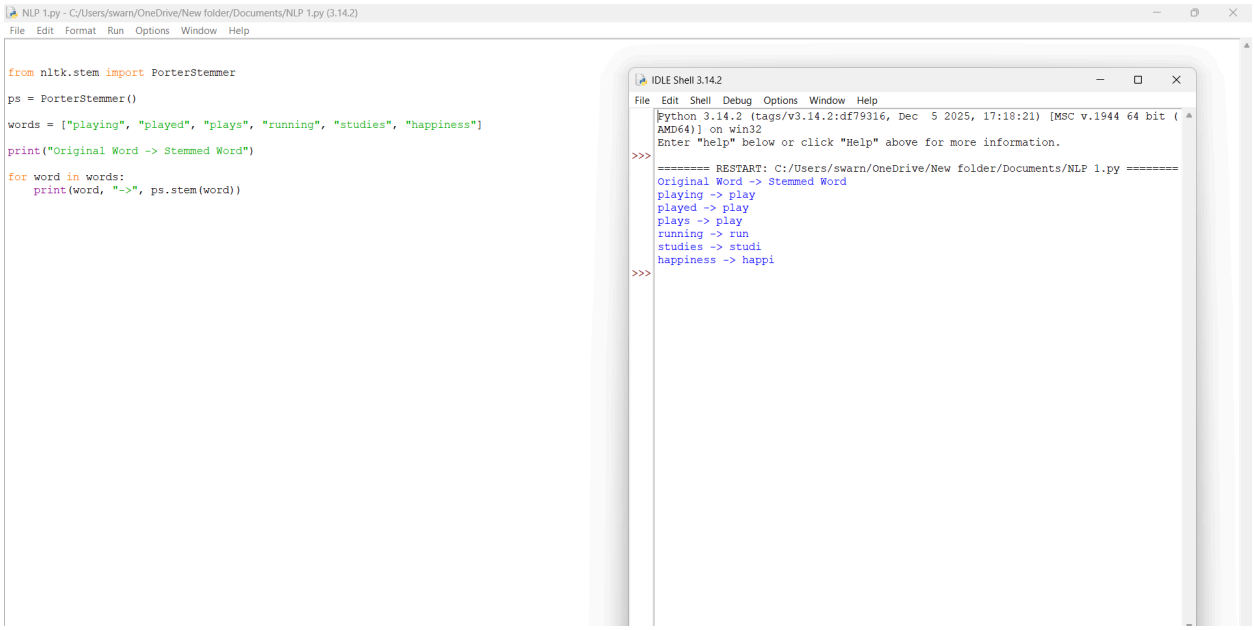

ps = PorterStemmer()


words = ["playing", "played", "plays", "running", "studies", "happiness"]


print("Original Word -> Stemmed Word")


for word in words:

    print(word, "->", ps.stem(word))
```



Result

Thus, the Porter Stemmer algorithm was implemented successfully using the NLTK library to obtain the stemmed forms of the given words.